BCS-069	Introduction to Internet of Things (IoT)	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	To learn about basic of IoT
CO2	Understand domain specific applications
CO3	Identify various protocols for IoT
CO4	To learn various M2M and IoT architectures
CO5	Understand various challenges of IoT

Module 1

Introduction to IoT: Defining IoT, Characteristics of IoT, Physical design of IoT, Logical design of IoT, Functional blocks of IoT, Communication models & APIs, Elements of an IoT ecosystem.
IoT APPLICATIONS - Home automation, Future Factory Concepts, Smart Objects, and Smart Applications. Study of existing IoT platforms /middleware.

Module 2

IoT Protocols: Protocol Standardization for IoT, M2M and WSN Protocols – SCADA and RFID Protocols, Issues with IoT Standardization, Protocols- IEEE802.15.4–BACNet Protocol– Modbus – KNX – Zigbee– Network layer – APS layer – Security
IoT Reference Architecture- Introduction, Functional View, Information View, Deployment and Operational View, Other Relevant architectural views. Real-World Design Constraints- Introduction, Technical Design constraints-hardware is popular again, Data representation and visualization, Interaction and remote control.

Module 3

Challenges in IoT: Design challenges, Development challenges, Security challenges, other challenges.
Tools: Introduction to Python and Introduction to different IoT tools.

Textbook:

- Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos, David Boyle, “From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence”, 1st Edition, Academic Press, 2014.

Reference Books:

- Vijay Madisetti and Arshdeep Bahga, “Internet of Things (A Hands-on-Approach)”, 1stEdition, VPT, 2014.
- Francis daCosta, “Rethinking the Internet of Things: A Scalable Approach to Connecting Everything”, 1st Edition, Apress Publications, 2013
- Waltenegus Dargie,Christian Poellabauer, "Fundamentals of Wireless Sensor Networks: Theory and Practice"

Course Outcomes: After the completion of the course the student will be able to:

CO1	Understand fundamental concepts of Internet of Things (IoT)
CO2	Learn basic programming for implementation
CO3	Identify various devices, sensors and applications
CO4	Analyze various M2M and IoT architectures
CO5	Evaluate design issues in IoT applications